

EXHALE: Does starting from a Parker-wind initial condition speed up (and stabilize) convergence?

*We test whether seeding the time integration with a transonic (Parker-type) wind profile — in particular a hot Parker wind that also carries an elevated temperature and ionization, as a p-winds-style initial guess — reduces the number of iterations (and wall-clock) needed to reach the converged photoevaporative wind, compared with the default cold hydrostatic initial condition (IC). A new Hot Parker IC option was added to `set_IC.f90`; the benchmark runs the same planet (HD 209458b, H/He, spherical) from three ICs and compares convergence and the IC-vs-final profiles. **Bottom line:** the naive full-Parker-wind seed diverges (its isothermal density is globally inconsistent with the non-isothermal equilibrium and drains onto the base; §5–6); a corrected warm-seed IC that keeps the stable hydrostatic density and warms only the thermal/ionization/velocity overlay converges to the identical steady state (§7), but gives no speedup for this planet, because ATES's convergence is gated by hydrodynamic momentum-steadiness, not the thermal relaxation the seed shortcuts. The cold hydrostatic IC remains the recommended default.*

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1. Why the initial condition matters

ATES integrates the 1-D radiation–hydrodynamics to a *steady state*; the initial condition only sets where the relaxation starts, not the final answer. But it sets *how long* the relaxation takes. The cost has two parts:

1. **Hydrodynamic relaxation** — the velocity and density settle into the transonic outflow topology (subsonic base → sonic point → supersonic wind, with $\rho v r^2 = \text{const}$).
2. **Thermal/ionization relaxation** — the cold, nearly-neutral gas is photo-heated to $\sim 10^4$ K and ionized; the temperature and ionization profiles settle to radiative/advective balance. This is the *stiff* part, and historically the dominant cost (it motivated the semi-implicit energy solver; see `Update_EXHALE_solver`).

The current default IC (`set_IC.f90`) is an **isothermal hydrostatic atmosphere**: $\rho(r) = \rho_{bc} e^{b_{0,eff}(-\phi(r)+\phi(0))}$, a near-zero velocity seed $v = \frac{1}{2}(r-r_0)$, $T = T_0$ (cold base temperature), and gas that is *mostly neutral*. The wind must therefore launch from rest *and* heat/ionize from cold/neutral — both relaxations run from far away.

2. Parker / p-winds as an initial guess

A natural idea is to start from an analytic transonic wind. The isothermal **Parker wind** is the algebraic solution of the steady isothermal momentum equation in the gravitational (or Roche) potential,

$$\frac{1}{2}v^2 - c^2 \ln v - 2c^2 \ln r + \phi(r) = B, \quad (1)$$

with c^2 the (dimensionless) isothermal sound speed² and B fixed by the sonic point r_c where $d\phi/dr = 2c^2/r_c$; the wind takes the subsonic root for $r < r_c$ and the supersonic root for $r > r_c$, and $\rho = \dot{M}/(vr^2)$.

p-winds (dos Santos et al. 2022) is a popular open-source implementation of exactly this: an isothermal Parker wind plus a steady-state H/He ionization balance (and the He 2^3S triplet). Crucially, p-winds *prescribes* the temperature T and mass-loss rate $\dot{M}$ (free parameters fit to data); it does *not* solve the energy equation. Used as an ATES IC it would supply:

- the transonic $v(r)$, $\rho(r)$ topology (shortcuts hydrodynamic relaxation);
- **and the ionization structure** (shortcuts part of the ionization relaxation) — the key advantage over a bare cold Parker wind;
- at a **warm** temperature (right velocity scale in the heated region),

but it cannot supply the *non-isothermal* $T(r)$ of the real wind (cold base $T_0 \rightarrow \sim 10^4$ K plateau), and its $T, \dot{M}$ must be chosen.

Design choice. EXHALE already contains an isothermal Parker-wind solver (the Transonic IC option: Eq. 1 at the cold T_0 sound speed). Its limitation is that it is cold (wrong velocity scale) and neutral. We therefore capture the p-winds advantage *without an external dependency* by generalizing it to a **hot Parker IC**: the same Parker solve at a warm effective sound speed, plus a temperature and ionization ramp.

3. The hot-Parker IC in EXHALE

Implemented in `src/modules/init/set_IC.f90`, enabled by an optional `input.inp` line

Hot Parker IC: 10000 # effective wind temperature T_wind [K]

(`hot_parker_ic`, `T_wind_ic` in `parameters.f90`; parsed in `input_read.f90`; auto-enables `Force start`).

Velocity / density. The Parker solver `wind_profile` (now taking c^2 as an argument) is called with a *warm* sound speed

$$c_{\text{hot}}^2 = \frac{1 + \delta p_{\text{bc}}}{\rho_{\text{bc}}} \frac{T_{\text{wind}}}{T_0} \times 2, \quad (2)$$

where the factor 2 approximates the mean-molecular-weight halving when H is ionized ($\mu \rightarrow \mu/2$). $v(r)$ is the transonic root of Eq. 1 and $\rho(r) = \dot{M}/(vr^2)$ pinned to ρ_{bc} at the base. If the warm sonic point falls outside the domain the routine returns failure and `set_IC` falls back to the hydrostatic IC.

Temperature / ionization ramp. With $\xi(r) = \min(v/c_{\text{hot}}, 1)$ (0 at the subsonic base $\rightarrow 1$ in the supersonic wind),

$$T(r) = T_0 [1 + (T_{\text{wind}}/T_0 - 1)\xi], \quad (3)$$

$$x_{\text{HII}}(r) = \delta p_{\text{bc}} + (1 - \delta p_{\text{bc}})\xi, \quad x_{\text{HeII}}(r) = \xi, \quad (4)$$

and the pressure is kept thermodynamically consistent, $p = \rho c_{\text{base}}^2 (T/T_0)(1 + \xi)$, so that $p/\rho \rightarrow c_{\text{hot}}^2$ in the supersonic wind and $\rightarrow c_{\text{base}}^2$ at the cold base (matching the base BC). Metals are left neutral (they re-equilibrate per cell in one step). The base boundary condition ($T = T_0$, ρ_{bc}) is unchanged, so a thin base-reconciliation layer still relaxes.

IC dump. `set_IC` now writes the (dimensional) initial profile to `output/IC_dump.txt` (r, n, v, p, T) for any IC, so the IC can be plotted directly against the converged result.

4. Benchmark setup

Planet: HD 209458b-like hot Jupiter (the standard ATES test): $R_p = 1.38 R_J$, $M_p = 0.69 M_J$, $T_0 = 1320$ K, $a = 0.047$ AU, $M_* = 1.13 M_\odot$, power-law EUV+X-ray SED ($\log_{10} L_{\text{EUV}} = 29.3$), **H/He only** (no metals, for a clean and fast comparison), **spherical** domain to $10 R_p$. Solvers: the current defaults (semi-implicit energy, Brent T , analytic-Jacobian Newton).

Three initial conditions (same planet, same numerics):

1. hydrostatic — the default cold hydrostatic IC.
2. transonic — the cold (T_0) Parker IC (Transonic IC: True).
3. hot_parker — the warm Parker IC (Hot Parker IC: 10000).

Measured: iterations to convergence (the count at which `du/dtu` stall stops the run), wall-clock (OMP_NUM_THREADS=4), and the IC-vs-final profiles (T, v, ρ vs r). Files live in `ic_start_benchmark/` (one subdirectory per IC); the figures are produced by `ic_start_benchmark/ic_compare.ipynb`.

5. Results

The three runs are summarized in Table 1. The headline result is *negative and instructive*: neither Parker-wind start helped, and the warm (hot-Parker) start actively destabilized the integration.

Table 1: Three initial conditions, same planet (HD 209458b, H/He, spherical, $10R_p$), identical numerics (semi-implicit energy, Brent T , analytic-Jacobian Newton, OMP_NUM_THREADS=4).

IC requested	IC actually used	Steps	Wall [s]	Outcome
hydrostatic	hydrostatic (cold)	95 274	1828.8	converged ($du=2.0e-2$, $dtu=6.2e-7$)
transonic	hydrostatic (fallback)	95 274	1836.7	identical to hydrostatic
hot_parker	warm Parker, $T_{\text{wind}}=10^4$ K	30 421*	604.8*	diverged — NaN at base

* The hot-Parker step count and wall-clock reflect an *early numerical breakdown*, *not* convergence: the run terminated on a NaN-corrupted state ($du=5.1e+2$, $dtu=1.3e-2$; $T > 10^{40}$ K at $r \approx 1.0002$). It must not be read as a speedup.

transonic $\equiv$ hydrostatic. At the cold base sound speed $c_0^2 = kT_0/\mu$, the Parker sonic point r_c (where $d\phi/dr = 2c_0^2/r_c$) falls *outside* the computational domain for HD 209458b — there is no interior transonic root. `set_IC` detects this (“Transonic IC requested but no interior sonic point; falling back to hydrostatic IC”) and uses the hydrostatic IC, so the two runs are byte-identical. A *cold* Parker IC offers nothing here.

hot_parker diverges at the base. The warm-Parker IC itself is smooth and physical (Fig. 1, dashed blue): a subsonic warm base ($v \approx 0.7$ km s $^{-1}$, $T \approx 1840$ K, $n \approx 6 \times 10^{12}$) ramping monotonically to a supersonic 10^4 K wind ($v \approx 23$ km s $^{-1}$ at $10R_p$), with no NaN in the IC dump. But during time integration the base *cannot reconcile* the warm, outflowing interior with the cold fixed base boundary condition ($T = T_0 = 1320$ K, $\rho = \rho_{bc}$): within a few cells of $r = 1$ the base develops an *inflow* (“breathing”, final $v \approx -1.1 \times 10^5$ cm s $^{-1}$), the cell-by-cell energy solve latches onto the *spurious hot root*, and T runs away to $> 10^{40}$ K. This is the exact breathing-base hot-root pathology documented for the `_adv` post-processor (the option-(c) guard; see `Update_EXHALE_solver`) — but here it strikes the *main* time integration, where no such guard exists.

6. Root-cause analysis: why the warm start diverges

A time-marching code *should* relax any initial condition to the same steady state, so a divergence is not inevitable — it points to a specific, identifiable mechanism. We instrumented the base cells (an env-gated diagnostic, `ATES_DIAG_BASE=1`, dumping r, n, v, T , heat, cool for the lowest cells over the first 500 steps; run with 60 OpenMP threads) and traced the exact onset (Fig. 4). The causal chain is:

- Inverted base pressure gradient (built into the IC+BC pairing).** The lower boundary condition (`Apply_BC.f90`) pins the ghost cells to the *cold* base state: $\rho_{\text{ghost}} = \rho_{bc}$, $p_{\text{ghost}} = 1 + \delta p_{bc}$ (hence $T_{\text{ghost}} = T_0 = 1320$ K), and $v_{\text{ghost}} = \max(v_1, 0)$. This is exactly consistent with a cold hydrostatic IC. But the warm IC places the first *interior* cells at the warm pressure $p \approx 1.35$ ($T \approx 1840$ K, $\sim 1.4\times$ the cold value at the same density). So $p_{\text{interior}} > p_{\text{ghost}}$ — an *inverted* pressure gradient at the base, whose force points *inward* and adds to gravity.
- The base flips to inflow within 3 steps** (Fig. 4a). The base velocity reverses from the IC outflow ($+0.4$ km s $^{-1}$) to inflow (-0.09 km s $^{-1}$ at step 3) and then grows *monotonically* more negative, reaching -2.5 km s $^{-1}$ by step 480.

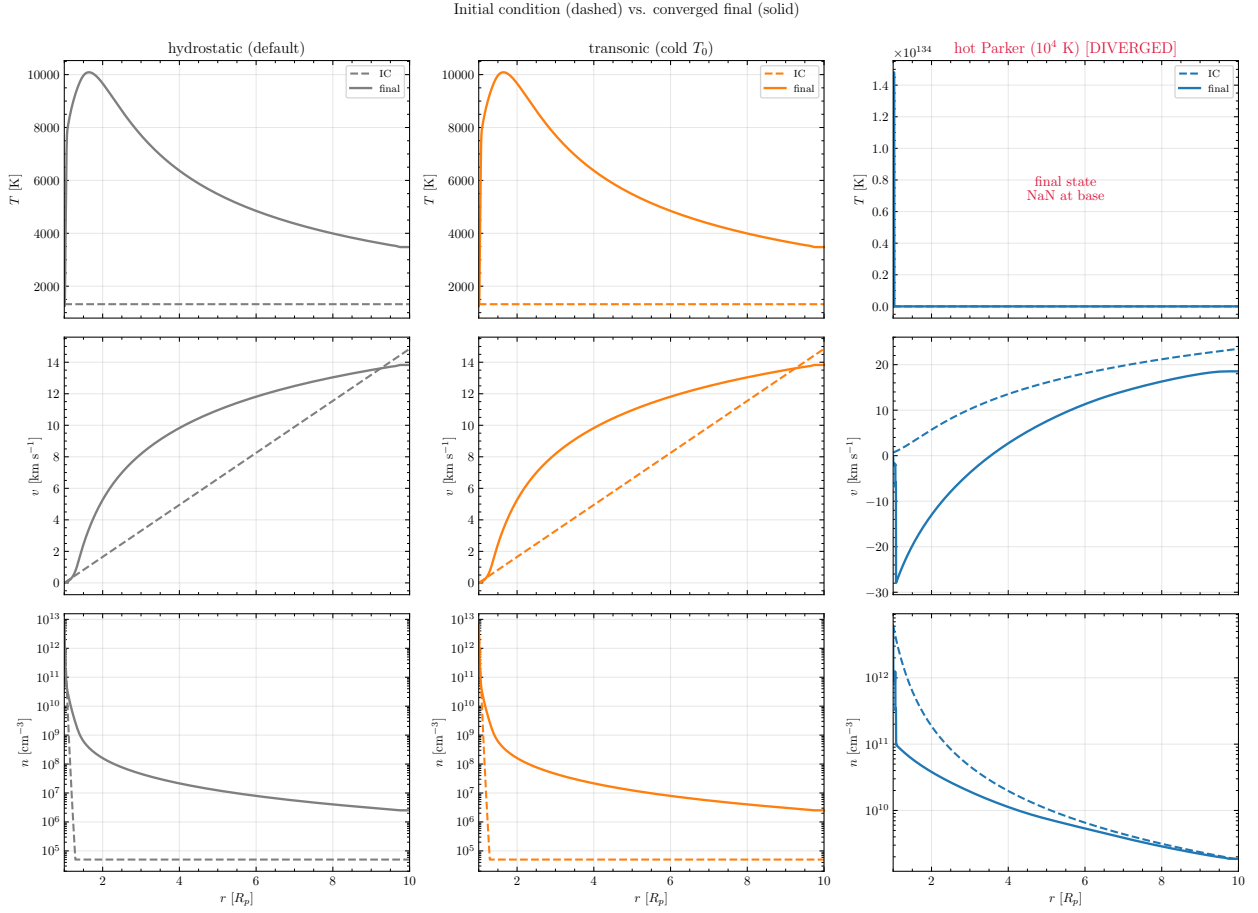


Figure 1: Initial condition (dashed) vs. converged final (solid) for each IC: temperature (top), velocity (middle), density (bottom). **hydrostatic** and **transonic** (the latter identical, having fallen back) relax from the flat cold IC to the peaked $\sim 10^4$ K transonic wind. **hot_parker** starts from a sane warm-wind IC but its final state diverges (note the 10^{134} K temperature axis and the “NaN at base” annotation; the final velocity turns negative — base inflow).

The BC’s $v_{\text{ghost}} = \max(v_1, 0)$ clamp acts as a **one-way valve**: it permits the inflow but supplies *zero* outflow to relieve it, so the base never re-establishes the wind.

3. **The base compresses, it does not drain** (Fig. 4b). The inflow piles gas up: n_{base} rises from 5.4×10^{12} to 9.8×10^{12} over 500 steps. (This corrects the naive “density-collapse” guess — the failure is compression, not evacuation.)
4. **A huge net heating with no advective escape** (Fig. 4d). The base is photoionization-dominated: heat/cool ≈ 2000 . In a *converged* wind this enormous net heating is carried off by the *outflow* (advection + PdV expansion cooling). An *inflowing*, compressing base has no such escape — PdV now does positive (compressional) work and advection imports energy — so the net heat accumulates.
5. **Slow T climb, then runaway over the cooling peak** (Fig. 4c). T_{base} rises steadily ($1675 \rightarrow 2710$ K over 500 steps); it has *not* yet reached the cooling peak at step 500 — the full runaway comes much later ($T \rightarrow 10^{40} \rightarrow \text{NaN}$ near step 30000). The trigger is in the energy solver: in `energy_semi_implicit.f90` the implicit factor

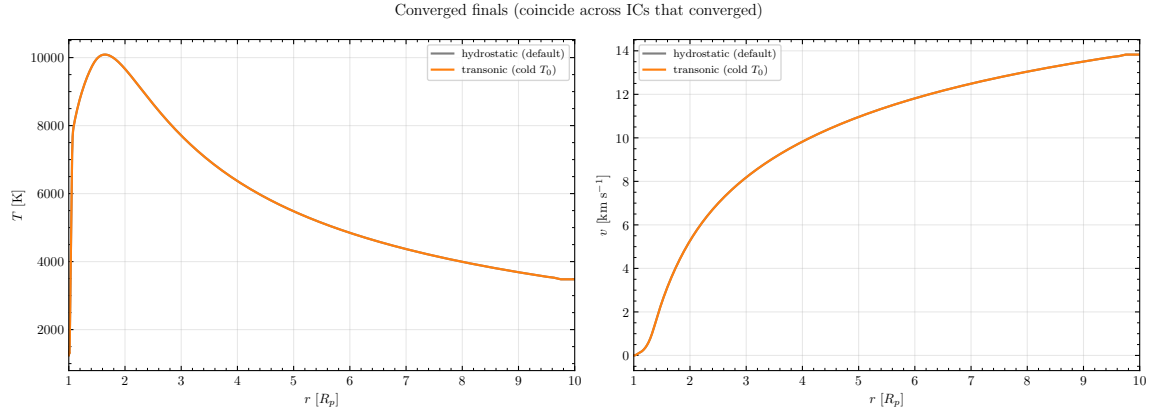


Figure 2: Converged final temperature and velocity for the ICs that converged (hydrostatic and transonic): they coincide, confirming the IC affects only convergence cost, not the steady state. The diverged hot-Parker run is excluded.

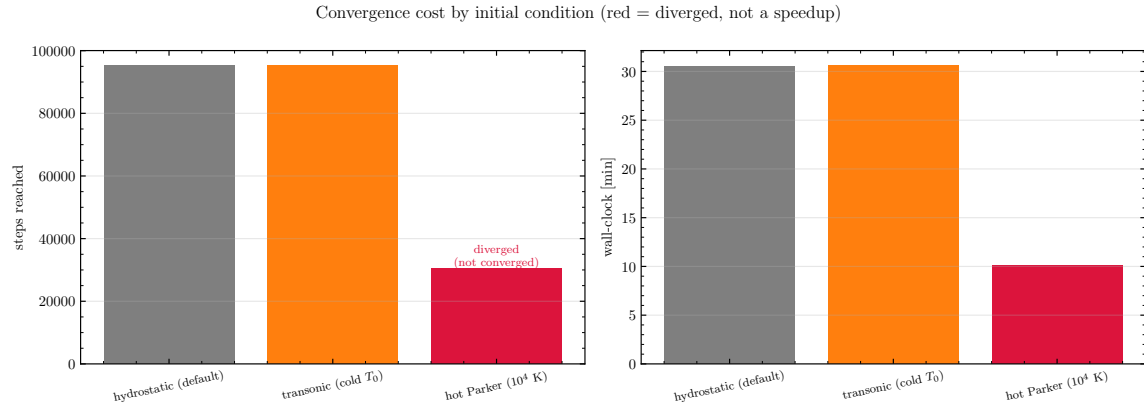


Figure 3: Steps and wall-clock per IC. The hot-Parker bars (red) are an early numerical breakdown, *not* a speedup — the run never converged.

is $dF/dT = 1 + c_{\text{factor}} \max(0, dC/dT)$, which is unconditionally stable *only on the rising branch* of the cooling curve. Once a base cell is pushed past the cooling peak ($\sim 1.5\text{--}2 \times 10^4$ K for photoionized H/He, where the neutral fraction collapses and the cooling rate *falls* with T , i.e. $dC/dT \leq 0$), the $\max(0, \cdot)$ disengages the safeguard, $dF/dT \rightarrow 1$, the update becomes *fully explicit with no damping*, and the temperature runs away.

So the divergence is the product of two design assumptions, each individually reasonable: a base BC *hard-wired to the cold state*, and an energy solver whose unconditional stability holds *only on the rising branch* of the cooling curve. The cold hydrostatic IC sits in the basin of attraction of both — it matches the cold BC (no inverted gradient, no base inflow) *and* it approaches equilibrium from below (never visiting the $dC/dT \leq 0$ branch). The warm Parker IC violates both at once.

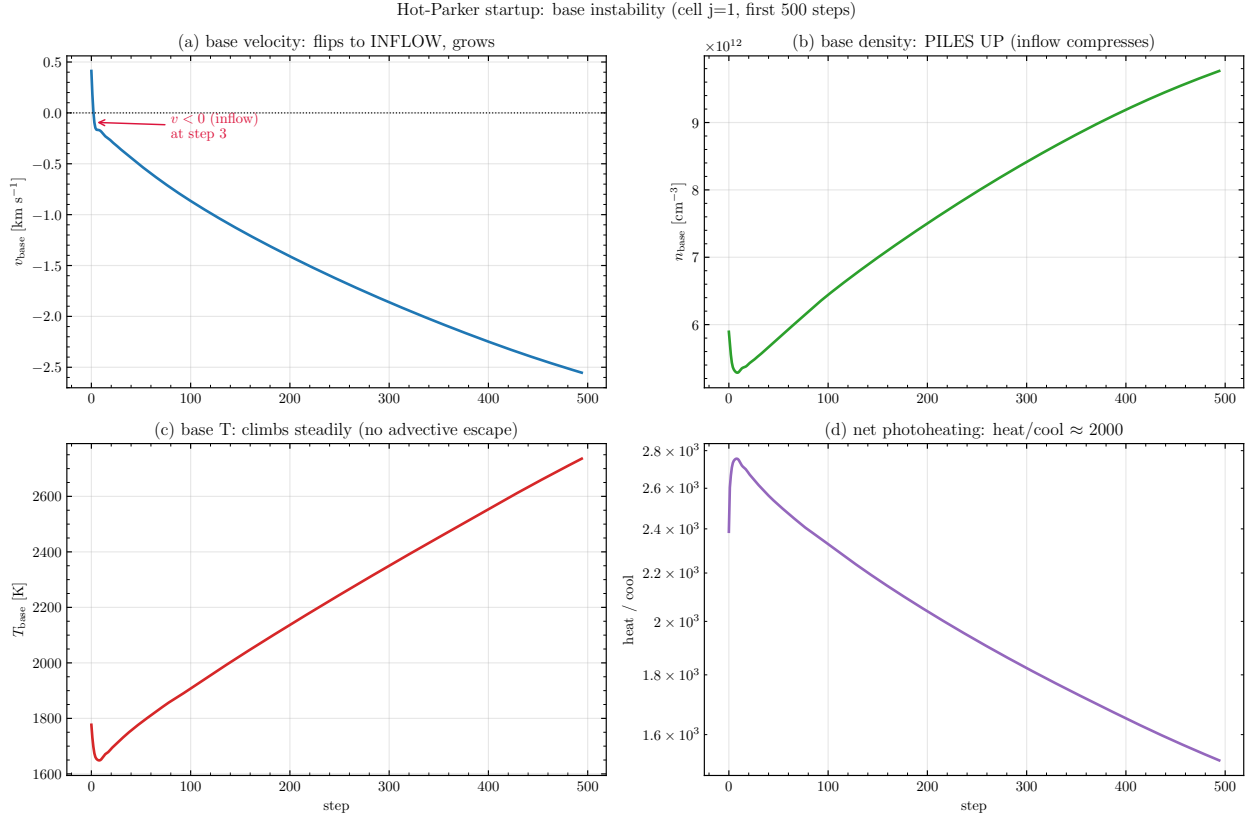


Figure 4: Instrumented hot-Parker startup (base cell $j=1$, first 500 steps, 60 threads). (a) the base velocity flips from the IC outflow to *inflow* by step 3 and grows monotonically more negative (the one-way-valve base BC cannot relieve it); (b) the base *compresses* (density piles up) rather than draining; (c) T climbs steadily with no advective escape; (d) the base carries a near-constant net photoheating heat/cool ≈ 2000 . The cell has not yet crossed the cooling peak at step 500; once it does, the semi-implicit energy safeguard disengages and T runs away to NaN ($\sim$ step 30000).

7. Making the warm start converge (implemented fix)

The root-cause analysis (§6) named two candidate triggers — the cold-pinned base BC and the energy-solver branch. Acting on them showed that the *decisive* one is neither: it is the **IC density**.

First attempt (insufficient): cold-base window on the full Parker IC. Forcing the base layer cold and static with a spatial smoothstep window (cold below a transition radius r_{tr} , warm Parker above) made the IC *start* correct (base $T = T_0$, $v \approx 0$, $p = 1 + \delta p_{bc}$), but the run still diverged: the base inflow *re-developed dynamically*, growing monotonically to $-2.7 \times 10^5 \text{ cm s}^{-1}$ over the first 500 steps. So the inverted IC gradient was not the sustaining cause.

The diagnostic that settled it. Running the *cold hydrostatic* IC with the same base instrumentation revealed that it, too, breathes inward — the base velocity is negative in 494 of the first 500 steps. Base breathing is therefore *normal*. The difference is the dynamics: the cold-hydrostatic inflow is **self-limiting** (it peaks at $-2.6 \times 10^4 \text{ cm s}^{-1}$ near step 240 and then *recedes*), whereas the warm-Parker inflow is **self-amplifying** ($\sim 10\times$ larger, monotonic). The culprit is the *isothermal-Parker density*: it is globally inconsistent with ATES’s non-isothermal equilibrium — the hot, low-density

Parker column is not supported against gravity at the cold base and drains onto it, feeding the inflow.

The fix that works: keep the hydrostatic density, warm only the overlay. The working warm-seed IC (now what Hot Parker IC builds) keeps the **stable cold hydrostatic density** — the profile the cold IC proves is balanced — and overlays on it only (i) a warm temperature + H/He ionization ramp (which shortcuts the stiff thermal/ionization relaxation) and (ii) the Parker *velocity* as a transonic head-start, both anchored to the cold static base by the smoothstep window ($x_i = 0$ below $r = 1$, $\rightarrow 1$ at $r \geq \text{hp_base_rtr}$). The Parker solve is used for the velocity only; its density is discarded. With this, the base inflow becomes self-limiting again (peak -2.8×10^4 , then recedes — indistinguishable from the cold-hydrostatic base), and the run converges with no NaN.

Result: it converges to the same state, but does not speed up. A clean head-to-head (same planet, 60 OpenMP threads, both to convergence) gives Table 2: the warm-seed IC converges to the *identical* steady state ($\log_{10} \dot{M} = 11.52$ for both) in essentially the *same* cost (93068 vs 95274 steps, -2.3% ; wall-clock $+0.5\%$ — within noise). Figure 5a shows why: although the warm seed’s early transient is markedly *gentler* (its du peaks near 10^3 versus $\sim 10^6$ for the cold start), the two convergence histories merge after ~ 15000 steps and the long slowly-decaying tail — which sets the total step count — is identical. ATES’s stopping criterion is the **hydrodynamic momentum-steadiness** test ($du < 2 \times 10^{-2}$, i.e. $\rho v r^2 = \text{const}$), and the warm seed accelerates the *thermal* relaxation, not that hydrodynamic one. So for this (momentum-gated) planet the warm start is a correctness-neutral option, not a performance win.

Table 2: Cold hydrostatic vs. the fixed warm-seed IC (Hot Parker IC), both run to convergence at 60 OpenMP threads. Same converged $\dot{M}$; no meaningful speedup.

IC	Steps	Wall [s]	Converged	$\log_{10} \dot{M}$
cold hydrostatic	95 274	1851.1	yes ($du=2.0\text{e-}2$)	11.52
warm-seed (fixed)	93 068	1860.9	yes, no NaN	11.52

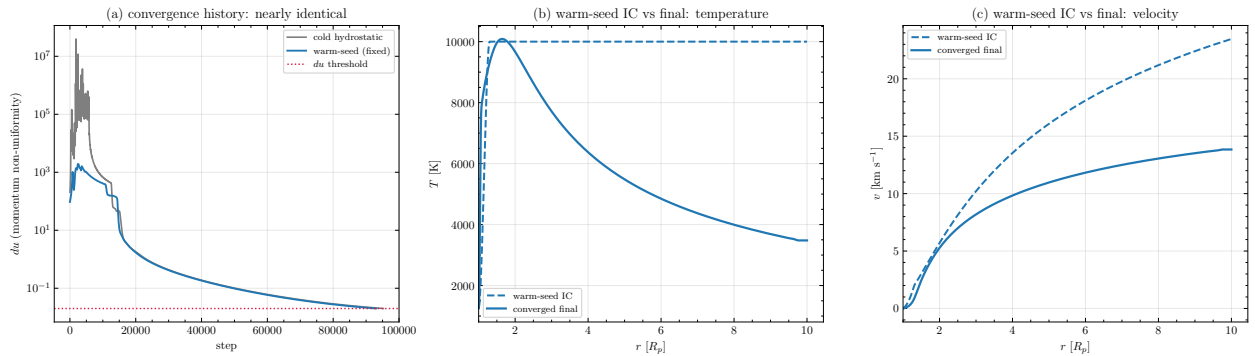


Figure 5: The fixed warm-seed IC. (a) du convergence history: the warm seed’s early transient is gentler (peak $\sim 10^3$ vs $\sim 10^6$), but the histories merge and the step-count-setting tail is identical — convergence is momentum-gated. (b,c) the warm-seed IC (dashed: already warm/ionized, with a Parker velocity head-start) relaxes to the correct non-isothermal converged profile (solid).

8. Discussion and recommendation

Does a Parker-wind start speed up ATES? No — and that is consistent with how ATES converges. ATES is a *time-marching* code: it relaxes any initial condition to the same steady state, and the IC only sets the cost of getting there. The two relaxation channels (§1) respond differently to a Parker seed:

- The **hydrodynamic** channel *would* benefit from a transonic seed — but only if a consistent transonic IC can be imposed at the base. For this planet the cold Parker has no interior sonic point, and the warm Parker conflicts with the base BC.
- The **thermal/ionization** channel is the stiff, dominant cost. A warm/ionized seed could shortcut it — but only if it does not break the base. It broke the base.

Why the warm seed was fragile — and what actually fixed it. The original divergence (§6) was real and proceeded exactly as traced (inverted base gradient → base inflow → no advective escape → slow heating → runaway past the cooling peak). But §7 showed that the *sustaining* cause was deeper than the inverted IC gradient: even with the base started cold, the inflow re-developed, because the isothermal-Parker *density* is globally inconsistent with the non-isothermal equilibrium and drains onto the base. The decisive fix was therefore to discard the Parker density and keep the stable cold hydrostatic density, warming only the temperature/ionization/velocity overlay. That converges to the correct steady state (Table 2).

Recommendation. Keep the **cold hydrostatic IC as the default**: even the *fixed* warm seed gives no speedup for this momentum-gated planet (§7), so there is no performance reason to switch. The `Hot Parker IC` keyword now builds the *validated* warm-seed IC (stable hydrostatic density + warm overlay, cold-base smoothstep window `hp_base_rtr`); it is a correct, converging option rather than an experimental one, and may help on a problem whose convergence is *thermally* gated rather than momentum-gated (e.g. a denser, more strongly-cooled base where the thermal relaxation, not the wind, sets the step count). Two further, independent improvements remain available:

1. **A characteristic / pressure-gradient base BC** (instead of the hard cold-pinned ρ, p with $v = \max(v_1, 0)$) would let the base breathe physically and widen the basin of attraction for non-hydrostatic starts generally.
2. **Make the energy solver unconditionally stable on both branches** — use a true backward-Euler step on the full cooling (or take $|dC/dT|$ rather than $\max(0, dC/dT)$ in `energy_semi_implicit.f90`), so a cell pushed past the cooling peak during *any* transient cannot run away. This is the same spirit as the option-(c) breathing-base guard already in the `_adv` post-processor, applied to the main integration. (Not required for the warm-seed IC above, which no longer reaches the peak, but a worthwhile robustness margin.)

The instrumentation used here is retained (`env-gated ATES_DIAG_BASE=1`; zero effect when unset) so any future base-stability work can re-trace the startup directly.

Contrast with relaxation BVP codes. This negative result actually highlights an ATES strength. Steady-state relaxation codes such as `wind-ae` (Numerical Recipes `solvde` two-point BVP) *require* a transonic initial guess close to the

target solution — their own documentation warns the relaxation “is incredibly sensitive to being given a good initial guess ... if the guess is too far from the goal, the relaxation method will not converge,” and they ship parameter “rampers” to walk a known solution to a new one. ATES needs no such guess: it converges from a generic cold hydrostatic atmosphere precisely because it marches in time. The cost is that the stiff thermal relaxation is slow; the benefit is robustness. The Parker-start experiment shows that trying to buy the BVP-code’s head start, without their careful base treatment, trades that robustness away.